ETM — Execution Trace Module

Standard: Execution Layer — Enforcement Standard

Category: Governance & Enforcement

Subcategory: Execution & Enforcement Systems

Type: Traceability & Audit Module

Version: 1.0

Status: Canonical · Open

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Canonical Definition

Execution Trace defines the conditions under which all actions within a system are recorded and reconstructable.

An action is valid only if it is fully traceable, time-linked, and attributable to a defined execution context.

If execution cannot be traced, it is invalid.

Purpose

ETM ensures that every action:

- is recorded
- can be reconstructed
- is linked to time and context

It creates a verifiable execution history.

Core Principle

If it cannot be traced, it did not validly occur.

MIF — Minimum Implementation Framework

A system implementing ETM must:

- record executing entity
- record timestamp
- record input conditions
- record execution context
- enable reconstruction of execution

Core Conditions

- every action must be recorded
- execution must be time-linked
- execution must be attributable

- execution must be reconstructable

Failure Condition

Traceability is invalid if:

- execution is not recorded
- timestamp is missing
- actor cannot be identified
- context is incomplete
- reconstruction is not possible

Use Case 1 — Untraceable System Action

Scenario

A system performs an action, but no full record exists.

Without ETM

- action occurs
- no traceability
- investigation fails

With ETM

- action is recorded
- full execution path is available

Result

System behavior is verifiable.

Use Case 2 — Financial Transaction Audit

Scenario

A transaction must be audited after execution.

Without ETM

- partial logs exist
- key data is missing

With ETM

- full trace is available
- transaction can be reconstructed

Result

Audit becomes reliable and complete.

Closing Statement

Execution without traceability is not valid execution.

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